

Lévy-Score-Based Monte Carlo Additional Experiments

June 29, 2026

Abstract

We study the problem of drawing samples from a Boltzmann distribution $\pi(x) \propto \exp(-2V(x)/\sigma^2)$ on $\mathbb{R}^d$ whose mass is concentrated in several well-separated modes. Gradient-based Markov chain Monte Carlo (MCMC) mixes well *within* a basin but crosses inter-mode barriers only at a rate that decays exponentially in barrier height and dimension. *Lévy-Score-Based Monte Carlo* (LSB-MC) augments overdamped Langevin dynamics with a compound-Poisson jump process and a drift correction—the *Lévy score*—chosen so that the jump-diffusion retains π as its invariant law. We give the generator of the resulting process and a short proof that the Lévy-score drift makes π stationary (Proposition 1), state the discretization and quadrature actually used, and define every evaluation metric precisely. Empirically we benchmark LSB-MC against six samplers, including the two strongest generic multimodal methods (parallel tempering and Hamiltonian Monte Carlo), under two scaling axes—mode count $K \in \{10, 20, 40, 80\}$ and dimension $d \in \{8, 16, 32, 64\}$ —and on a Bayesian Gaussian-mixture posterior whose $K!$ modes arise from an exact label-permutation symmetry. All runs are single-GPU, seed-replicated with error bars, and charged for compute (wall-clock and gradient/potential evaluations).

1 Introduction

Let $\pi(x) \propto e^{-2V(x)/\sigma^2}$ be a target density on $\mathbb{R}^d$. Many tasks in Bayesian inference and statistical physics reduce to estimating $\mathbb{E}_\pi[\varphi]$ for test functions φ , which Monte Carlo addresses by producing samples $x_1, \dots, x_n$ with empirical law $\hat{\pi}_n = \frac{1}{n} \sum_i \delta_{x_i} \approx \pi$. The difficulty addressed here is *multimodality*: π has several modes separated by regions of low density. For a diffusion-based sampler the expected time to move between two modes separated by a barrier of height Δ scales (Kramers/Arrhenius) like $\exp(2\Delta/\sigma^2)$, so for tall barriers or many modes a chain initialised in one basin is, in practice, absorbed there.

Intuition. Picture π as a landscape whose deep valleys are the likely outcomes. Sampling drops many walkers and lets them move roughly downhill with noise, so the final crowd is densest in the deepest valleys. If the valleys are walled off from each other, walkers that only step locally stay in the valley they started in and never report the others—so the crowd misrepresents π .

Existing remedies and their ceiling. Tempering-based methods flatten the barriers in auxiliary high-temperature chains and exchange states downward; parallel tempering (PT) [1, 2] is the strongest general-purpose instance. Hamiltonian Monte Carlo (HMC) [3, 4] makes long gradient-guided proposals. Both reduce, but do not remove, the exponential barrier cost: PT needs an increasingly fine temperature ladder as d and the mode count grow, and HMC trajectories still cannot reliably traverse tall barriers.

The LSB-MC idea. Rather than crossing barriers by chance, LSB-MC lets particles take occasional large jumps drawn from a fixed, target-informed *jump bank*, and adds a drift correction (the Lévy score) that exactly offsets the bias such jumps would introduce, preserving π as the invariant law. Section 3 makes this precise; the correction is the mechanism, not a tunable add-on (Remark 1).

Contributions. (i) We give a self-contained continuous-time construction and an invariance proof for the Lévy-score correction (§3). (ii) We benchmark against the strong baselines PT and HMC, not only the weak ULA/MALA/FLMC set (§4). (iii) We report interpretable, formally defined metrics—mode coverage, mode-weight KL, effective mode count (EMC)—alongside W_2 /MMD (§6). (iv) We quantify scaling in mode count (§7) and dimension (§8), and study a genuine Bayesian posterior whose modes follow from a symmetry (§9). (v) We account for compute fairly (§10). The next page summarises the findings with live numbers.

2 Results at a glance

Across all three studies the picture is the same. The local samplers (ULA, MALA, FLMC, ULD, HMC) stay trapped near wherever they were started and never discover the rest of the distribution; parallel tempering (PT), the strongest generic baseline, recovers partial coverage but degrades steadily as the problem grows; and **LSB-MC reaches essentially the whole distribution**, attaining the best mode coverage and EMC and the smallest mode-weight KL, typically sitting at the finite-sample reference floor.

Discovering many modes (MoG). With $K=40$ equally weighted Gaussian modes and all particles started near one of them, LSB-MC covers 1.00 of the modes with EMC 0.94 (1.0 is perfect), while the best baseline (PT) reaches only coverage 0.23 and EMC 0.07; the corresponding Wasserstein distance is 10.00 for LSB-MC versus 33.85 for PT.

Scaling to high dimension (ManyWell). At $d=64$ (32 independent double wells), LSB-MC keeps a count-EMC of 0.96 and a per-block marginal KL of 0.001, i.e. each well is filled in almost exactly the right proportion, whereas PT and HMC have collapsed to count-EMC ≈ 0.00 ; the sliced-Wasserstein distance is 0.18 for LSB-MC versus 1.41 for PT.

A real Bayesian model (label switching). The posterior of a 3-component Gaussian mixture has six identical, relabelled copies. Every baseline finds only the one copy it started in (coverage 0.17, 1 of 6 modes, EMC 0.011), whereas LSB-MC populates all six uniformly (coverage 1.00, EMC 0.998) using a jump bank read directly off the label-permutation symmetry.

Compute is accounted for fairly (wall-clock time and gradient evaluations, with PT and HMC charged their extra work and LSB-MC charged its Lévy-score quadrature), so the advantage is not an artefact of spending more compute — though, as Section 10 shows, that quadrature does add a real and problem-dependent per-step cost.

3 Sampling dynamics and the Lévy-score correction

Setup and notation. Throughout, $\pi(x) = Z^{-1}e^{-2V(x)/\sigma^2}$ with $Z = \int e^{-2V/\sigma^2} dx < \infty$, and we fix $\sigma^2 = 2$ so that $\pi \propto e^{-V}$, the force is $-\nabla V$, and the diffusive noise has scale $\sigma\sqrt{\Delta t}$. We write $\nabla \cdot$ for divergence and Δ for the Laplacian.

Assumption 1. $V \in C^2(\mathbb{R}^d)$ is confining ($e^{-2V/\sigma^2} \in L^1$), and the jump bank is a finite measure $\nu(dr) = \sum_{j=1}^J \lambda_j \delta_{r_j}$ on $\mathbb{R}^d$ with total rate $\lambda = \sum_j \lambda_j < \infty$. These guarantee the generators below act on C_c^2 and the manipulations are justified by dominated convergence.

3.1 Overdamped Langevin diffusion

The reference dynamics is $dX_t = -\nabla V(X_t) dt + \sigma dB_t$ with generator and Fokker–Planck adjoint

$$\mathcal{L}_0 f = -\nabla V \cdot \nabla f + \frac{\sigma^2}{2} \Delta f, \quad \mathcal{L}_0^* \rho = \nabla \cdot (\nabla V \rho) + \frac{\sigma^2}{2} \Delta \rho.$$

Since $\nabla \pi = -\frac{2}{\sigma^2} \pi \nabla V$, i.e. $\nabla V \pi = -\frac{\sigma^2}{2} \nabla \pi$, we get $\mathcal{L}_0^* \pi = \nabla \cdot (-\frac{\sigma^2}{2} \nabla \pi) + \frac{\sigma^2}{2} \Delta \pi = 0$: π is stationary (indeed reversible). What it lacks is *rate*: barrier crossings are exponentially rare, which is the failure mode of all the local samplers in this study.

3.2 Jump-augmented dynamics

Add finite-activity jumps with intensity ν and an auxiliary drift S . The resulting jump-diffusion has generator

$$\mathcal{A}f(x) = (-\nabla V(x) + S(x)) \cdot \nabla f(x) + \frac{\sigma^2}{2} \Delta f(x) + \int_{\mathbb{R}^d} [f(x+r) - f(x)] \nu(dr), \quad (1)$$

with Fokker–Planck adjoint

$$\mathcal{A}^* \rho(x) = \nabla \cdot ((\nabla V - S)\rho) + \frac{\sigma^2}{2} \Delta \rho + \int_{\mathbb{R}^d} [\rho(x-r) - \rho(x)] \nu(dr). \quad (2)$$

For $S \equiv 0$ we get $\mathcal{A}^* \pi = \mathcal{L}_0^* \pi + \int [\pi(\cdot - r) - \pi] \nu(dr) = \int [\pi(\cdot - r) - \pi] \nu(dr)$, which is generically nonzero: bare jumps destroy stationarity.

3.3 The Lévy score and exact invariance

Define the *Lévy score*

$$S_L(x) = -\int_0^1 \int_{\mathbb{R}^d} r \exp\left(-\frac{2}{\sigma^2}(V(x-\theta r) - V(x))\right) \nu(dr) d\theta = -\frac{1}{\pi(x)} \int_{\mathbb{R}^d} \int_0^1 r \pi(x-\theta r) d\theta \nu(dr), \quad (3)$$

where the second equality uses $e^{-\frac{2}{\sigma^2}(V(x-\theta r) - V(x))} = \pi(x-\theta r)/\pi(x)$.

Proposition 1 (Exact stationarity). *Under Assumption 1, with $S = S_L$ the jump-diffusion (1) admits π as a stationary density: $\mathcal{A}^* \pi = 0$.*

Proof. Because $\mathcal{L}_0^* \pi = 0$, by (2) it suffices to show

$$\nabla \cdot (S_L \pi)(x) = \int_{\mathbb{R}^d} [\pi(x-r) - \pi(x)] \nu(dr). \quad (4)$$

From (3), the flux is $S_L(x)\pi(x) = -\int_{\mathbb{R}^d} \int_0^1 r \pi(x-\theta r) d\theta \nu(dr)$. Fix r . Since $\nabla_x \cdot (r \pi(x-\theta r)) = r \cdot \nabla \pi(x-\theta r)$ and $\partial_\theta \pi(x-\theta r) = -r \cdot \nabla \pi(x-\theta r)$, we have $\nabla_x \cdot (r \pi(x-\theta r)) = -\partial_\theta \pi(x-\theta r)$. Integrating over $\theta \in [0, 1]$ telescopes:

$$\int_0^1 \nabla_x \cdot (r \pi(x-\theta r)) d\theta = -[\pi(x-r) - \pi(x)] = \pi(x) - \pi(x-r).$$

Therefore $\nabla \cdot (S_L \pi) = -\int_{\mathbb{R}^d} [\pi(x) - \pi(x-r)] \nu(dr) = \int_{\mathbb{R}^d} [\pi(x-r) - \pi(x)] \nu(dr)$, which is (4). Substituting into (2), the $-\nabla \cdot (S_L \pi)$ term cancels the jump source and $\mathcal{A}^* \pi = 0$. $\square$

Remark 1 (The correction is not an ablation). With $S \equiv 0$, $\mathcal{A}^*\pi = \int[\pi(\cdot - r) - \pi]\nu(dr) \neq 0$ in general, so the bare jump process is stationary for a *different* law. Hence “LSB-MC without the Lévy score” is not a degraded variant of the same sampler but a different (incorrect) one; we therefore do not include it as an ablation.

Remark 2 (Score interpretation). The integrand weight $\pi(x - \theta r)/\pi(x)$ is the density ratio along the chord from x to $x - r$; S_L is the bank-averaged displacement re-weighted by this ratio. Productive jumps (toward higher π) and unproductive ones are balanced so that the net probability flux induced by the jumps is exactly compensated.

Intuition. The jumps and the Lévy-score drift are a matched pair: a new movement plus the exact bookkeeping that keeps the stationary law equal to π . Equation (4) is that bookkeeping—the divergence of the corrective flux equals, pointwise, the net rate at which jumps move probability into each state.

3.4 Time discretization and quadrature

We integrate (1) by an explicit Euler–Maruyama step with a *tamed* drift for numerical stability under stiff V and post-jump transients. With $\text{tame}(v; \Delta t, c) = \Delta t v / (1 + (\Delta t/c)\|v\|)$, one iteration is

$$X^+ = X + \text{tame}(-\nabla V(X) + S_L(X); \Delta t, c) + \sigma\sqrt{\Delta t}\xi, \quad \xi \sim \mathcal{N}(0, I), \quad (5)$$

followed by the compound-Poisson jump: draw $N \sim \text{Poisson}(\lambda\Delta t)$ and set $X^{++} = X^+ + \sum_{m=1}^N r_{(m)}$ with each $r_{(m)} \sim \nu/\lambda$ (for the small Δt used here $N \in \{0, 1\}$ with high probability). The scheme is weakly consistent of order one with $\mathcal{A}$; π is exactly stationary only in the continuous-time limit, so a finite Δt leaves an $O(\Delta t)$ bias. Taming (parameter c) controls the explosion of the explicit scheme [6]; for stiff targets too large a c overshoots, so Δt and c are chosen jointly (see §9).

The Lévy score (3) requires the θ -integral, which we evaluate by n_θ -node Gauss–Legendre quadrature with nodes/weights (θ_q, w_q) :

$$S_L(x) \approx - \sum_{q=1}^{n_\theta} w_q \sum_{j=1}^J \lambda_j r_j \exp\left(-\frac{2}{\sigma^2}(V(x - \theta_q r_j) - V(x))\right),$$

at a cost of $n_\theta J$ potential evaluations per step (charged in §10). For ManyWell the integral is separable per coordinate and precomputed exactly on a fine grid, reducing the per-step cost to $O(1)$ lookups.

3.5 Jump banks

The only target-specific ingredient is ν , read off the geometry, never from the reference: (i) **ManyWell**: per double-well coordinate b , $\nu = \sum_b \frac{\lambda_b}{2}(\delta_{+re_b} + \delta_{-re_b})$ with $r \sim U[3.2, 3.8]$ bracketing the well separation; (ii) **MoG**: $\nu = \sum_{(i,j) \in \text{kNN}} \lambda \delta_{\mu_j - \mu_i}$ over a directed nearest-neighbour graph of the mixture centres; (iii) **Bayesian GMM**: $\nu = \sum_P \lambda \delta_{P\mu^* - \mu^*}$ over the label permutations P applied to a MAP estimate μ^* .

4 Samplers and baselines

All methods share the target $\pi \propto e^{-V}$ and are run as S independent particle batches (the seeds). We write $\xi \sim \mathcal{N}(0, I)$ for fresh noise.

ULA / MALA (local, gradient). ULA is (5) with $S_L \equiv 0$. MALA proposes $Y = X + \text{tame}(-\nabla V(X); \Delta t, c) + \sigma \sqrt{\Delta t} \xi$ and accepts with probability $\min\{1, \pi(Y)q(X | Y)/[\pi(X)q(Y | X)]\}$, where q is the Gaussian proposal kernel; this removes the discretization bias at the cost of a second gradient per step.

ULD (underdamped, BAOAB). With momentum p , inverse temperature $\beta = 2/\sigma^2$ and friction γ , the splitting integrator applies, per step, B: $p \leftarrow p - \frac{\Delta t}{2} \nabla V(x)$; A: $x \leftarrow x + \frac{\Delta t}{2} p$; O: $p \leftarrow e^{-\gamma \Delta t} p + \sqrt{(1 - e^{-2\gamma \Delta t})/\beta} \xi$; A; B. Its invariant law marginalises to π [5]; momentum reduces random-walk behaviour but the dynamics remain local.

FLMC (heavy-tailed). Langevin driven by isotropic symmetric α -stable noise ($\alpha = 1.5$): $X^+ = X + \text{tame}(-\nabla V(X); \Delta t, c) + c_\alpha \Delta t^{1/\alpha} L_\alpha$, L_α an $S\alpha S$ increment. The stationary law solves a fractional Fokker–Planck equation [7]; jumps are long but undirected.

HMC (strong baseline). On the augmented Hamiltonian $H(x, p) = V(x) + \frac{1}{2}\|p\|^2$, resample $p \sim \mathcal{N}(0, I)$, integrate L leapfrog steps of size ε , and accept with probability $\min\{1, e^{H(x, p) - H(x', p')}\}$. Each proposal costs $L+1$ gradient evaluations.

PT (strong baseline). A geometric ladder $1 = \beta_1 > \beta_2 > \dots > \beta_T \geq \beta_{\min}$ with replica t targeting $\pi_t \propto e^{-\beta_t V}$; each replica advances by a (tamed) MALA move, and every K_{swap} steps adjacent replicas $(t, t+1)$ propose a swap accepted with probability $\min\{1, \exp[(\beta_t - \beta_{t+1})(V(x_t) - V(x_{t+1}))]\}$. The cold chain ($\beta_1=1$) supplies the samples; cost is T kernel evaluations per step.

Step sizes for the Metropolis-based methods (MALA, HMC, PT) are tuned by short acceptance-rate pilots before the main runs.

5 Test problems

Each target exposes $\pi \propto e^{-V}$ and an exact or near-exact reference, so error is measured against ground truth.

ManyWell (high-dimensional, $d = 2n_{\text{blocks}}$). $V(x) = \sum_{b=1}^{n_{\text{blocks}}} [U(x_{2b-1}) + \frac{1}{2}x_{2b}^2]$ with the double well $U(u) = au + bu^2 + cu^4$, $(a, b, c) = (-0.5, -6, 1)$. The two wells per block are separated by ≈ 3.46 ; the bank magnitude $r \sim U[3.2, 3.8]$ brackets it. The target has $2^{n_{\text{blocks}}}$ modes; the reference is exact and i.i.d. via the per-coordinate inverse CDF. We sweep $d \in \{8, 16, 32, 64\}$.

MoG (mode discovery, $d = 2$). $\pi(x) = \frac{1}{K} \sum_{k=1}^K \mathcal{N}(x; \mu_k, I)$ with centres μ_k fixed (seed 0) spread over $[-R, R]^2$; all particles start near μ_1 , so the task is to discover and equally weight the remaining $K - 1$ modes. The reference is exact i.i.d. We sweep $K \in \{10, 20, 40, 80\}$.

Bayesian GMM label-switching ($d = 6$). For data $y_{1:n}$ from a $K=3$ -component, 2-D Gaussian mixture with unknown means $\mu = (\mu_1, \mu_2, \mu_3)$,

$$V(\mu) = - \sum_{i=1}^n \log \frac{1}{K} \sum_{k=1}^K \mathcal{N}(y_i; \mu_k, \sigma_y^2 I) + \sum_{k=1}^K \frac{\|\mu_k\|^2}{2\tau^2}.$$

For any label permutation P , $V(P\mu) = V(\mu)$, so the posterior has exactly $K!=6$ identical modes. The bank is the set of differences between permuted MAP modes; the reference symmetrises a long single-mode chain over all $K!$ permutations.

Intuition. Calling three clusters A, B, C fits the data exactly as well as calling them B, A, C : the labels are arbitrary. The posterior is therefore six mirror-image copies, and a correct sampler must visit all six with equal weight. Reporting only the copy one started in is a valid point estimate but a wrong posterior.

6 Evaluation metrics

Let $\hat{\pi}_n = \frac{1}{n} \sum_i \delta_{x_i}$ be the empirical sample law and π (or a large reference sample Q) the truth. We use two families: *distributional distances* (lower is better) and *mode-resolved scores* (coverage/EMC higher is better).

Maximum mean discrepancy. For a kernel k , $\text{MMD}_k^2(P, Q) = \mathbb{E}_{x, x' \sim P} k(x, x') + \mathbb{E}_{y, y' \sim Q} k(y, y') - 2 \mathbb{E}_{x \sim P, y \sim Q} k(x, y)$ [8]. We use a sum of Gaussian kernels $k(x, y) = \sum_{\ell} \exp(-\|x - y\|^2 / 2h_{\ell}^2)$ with h_{ℓ} multiples of the median pairwise distance, and report the (biased) $\text{MMD} = \sqrt{\text{MMD}^2}$.

Wasserstein-2. $W_2(P, Q) = (\inf_{\gamma \in \Pi(P, Q)} \int \|x - y\|^2 d\gamma)^{1/2}$. For MoG and the Bayesian model we compute the exact empirical W_2 by Hungarian assignment on the $n \times n$ cost matrix [10]; for high-dimensional ManyWell we use sliced- W_2 , $\text{SW}_2^2(P, Q) = \int_{\mathbb{S}^{d-1}} W_2^2(u_{\#}P, u_{\#}Q) du$, estimated over random projections with 1-D W_2 by sorting [9].

Mode-resolved scores. Assign each sample to its nearest mode (a Voronoi partition), giving empirical weights $\hat{p}_m$ against the target weights p_m^* . Then

$$\text{KL}(p^* \parallel \hat{p}) = \sum_m p_m^* \log \frac{p_m^*}{\max(\hat{p}_m, 1/n)}, \quad \text{EMC} = e^{-\text{KL}(p^* \parallel \hat{p})} \in (0, 1],$$

$$\text{coverage} = \frac{1}{M} \sum_m \mathbf{1}[\hat{p}_m > \max(1/n, 0.001 p_m^*)].$$

EMC=1 iff the mode weights match; coverage counts the fraction of modes actually populated.

ManyWell count metrics. The $2^{n_{\text{blocks}}}$ modes cannot be enumerated, so we use marginal statistics: the per-block marginal KL averaged over blocks, and the law of the number of occupied deep wells against its $\text{Binomial}(n_{\text{blocks}}, p_{\text{deep}})$ target (count-mode KL, and count-EMC= $e^{-\text{count-KL}}$), where p_{deep} is the exact deep-well marginal.

Protocol. Five seeds, realised as independent particle batches under a common PRNG stream; we report mean $\pm$ SE. Exactly one GPU is used (CUDA_VISIBLE_DEVICES pinned before initialisation); configs and environment are saved per run.

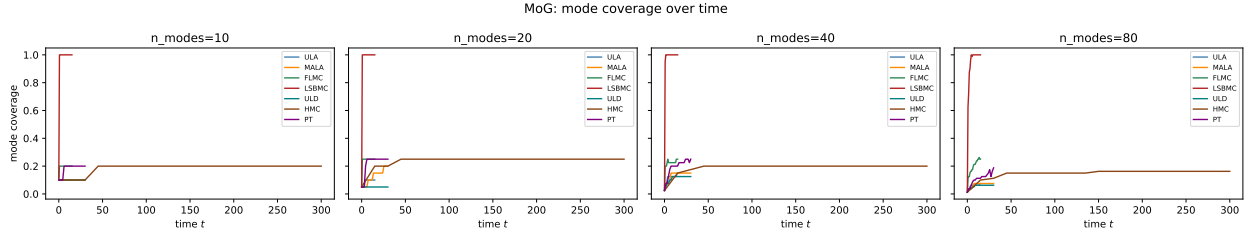


Figure 1: MoG mode coverage over sampling time (mean over seeds).

Table 1: MoG EMC across mode counts (mean $\pm$ SE).

Method	10	20	40	80
ULA	0.014 $\pm$ 0.000	0.020 $\pm$ 0.000	0.043 $\pm$ 0.000	0.065 $\pm$ 0.000
MALA	0.014 $\pm$ 0.000	0.020 $\pm$ 0.000	0.043 $\pm$ 0.000	0.065 $\pm$ 0.000
FLMC	0.021 $\pm$ 0.000	0.047 $\pm$ 0.000	0.072 $\pm$ 0.000	0.094 $\pm$ 0.001
LSBMC	0.924 $\pm$ 0.004	0.937 $\pm$ 0.004	0.943 $\pm$ 0.003	0.829 $\pm$ 0.005
ULD	0.014 $\pm$ 0.000	0.019	0.041 $\pm$ 0.000	0.063 $\pm$ 0.000
HMC	0.014 $\pm$ 0.000	0.029 $\pm$ 0.000	0.062 $\pm$ 0.000	0.094 $\pm$ 0.000
PT	0.021 $\pm$ 0.000	0.046 $\pm$ 0.000	0.070 $\pm$ 0.001	0.087 $\pm$ 0.000

7 Experiment 1: scaling in the number of modes (MoG)

Setup. K equally weighted Gaussian modes, all particles started near μ_1 ; a correct sampler reaches coverage and EMC near 1 with small W_2 /MMD. Because the start is concentrated, this isolates inter-mode *transport*.

Results. Table 1 reports EMC across the sweep and Table 2 the full metric set at $K=40$; Figure 1 tracks mode coverage over sampling time. At $K=40$, LSB-MC attains coverage 1.00 and EMC 0.94, versus PT’s coverage 0.23 / EMC 0.07 and HMC’s coverage 0.18; W_2 is 10.00 (LSB-MC) versus 33.85 (PT). LSB-MC’s EMC stays high across the sweep (0.92 at $K=10$, easing to 0.83 at $K=80$), while the local methods never leave μ_1 .

Discussion. The gap is qualitative, not incremental. W_2 is non-monotone in K for LSB-MC—an artefact of comparing point clouds under large separation—so coverage, EMC and MMD are the more reliable measures here; they agree that LSB-MC recovers the mixture while the baselines do not.

8 Experiment 2: scaling in dimension (ManyWell)

Setup. n_{blocks} independent double wells give $2^{n_{\text{blocks}}}$ modes; we sweep $d \in \{8, 16, 32, 64\}$. Independence makes the exact answer simple to state (each block deep a fixed fraction of the time), which lets us measure error cleanly despite the combinatorial mode count.

Results. Table 3 reports count-EMC across dimension and Table 4 the full metric set at $d=64$; Figure 2 tracks count-EMC over sampling time. At $d=64$, LSB-MC has count-EMC 0.96 and per-block marginal KL 0.001, versus PT’s count-EMC 0.00 and block KL 0.661; the sliced- W_2 is 0.18

Table 2: MoG results at $K = 40$ modes (mean $\pm$ SE over 5 seeds).

Method	W_2	MMD	cov.	EMC	KL	time(s)	grads
ULA	36.816 $\pm$ 0.421	0.642 $\pm$ 0.002	0.130 $\pm$ 0.005	0.043 $\pm$ 0.000	3.148 $\pm$ 0.004	0.279	1500
MALA	36.944 $\pm$ 0.434	0.643 $\pm$ 0.002	0.135 $\pm$ 0.006	0.043 $\pm$ 0.000	3.149 $\pm$ 0.007	0.917	3000
FLMC	33.397 $\pm$ 0.480	0.559 $\pm$ 0.002	0.230 $\pm$ 0.005	0.072 $\pm$ 0.000	2.630 $\pm$ 0.004	0.492	1500
LSBMC	10.003 $\pm$ 1.090	0.079 $\pm$ 0.007	1.000	0.943 $\pm$ 0.003	0.059 $\pm$ 0.004	55.418 $\pm$ 0.000	1500
ULD	37.204 $\pm$ 0.428	0.647 $\pm$ 0.002	0.125	0.041 $\pm$ 0.000	3.203 $\pm$ 0.004	0.264	1501
HMC	31.881 $\pm$ 0.348	0.544 $\pm$ 0.003	0.180 $\pm$ 0.005	0.062 $\pm$ 0.000	2.773 $\pm$ 0.004	2.026	16500
PT	33.847 $\pm$ 0.457	0.568 $\pm$ 0.004	0.235 $\pm$ 0.006	0.070 $\pm$ 0.001	2.664 $\pm$ 0.007	1.336	24000

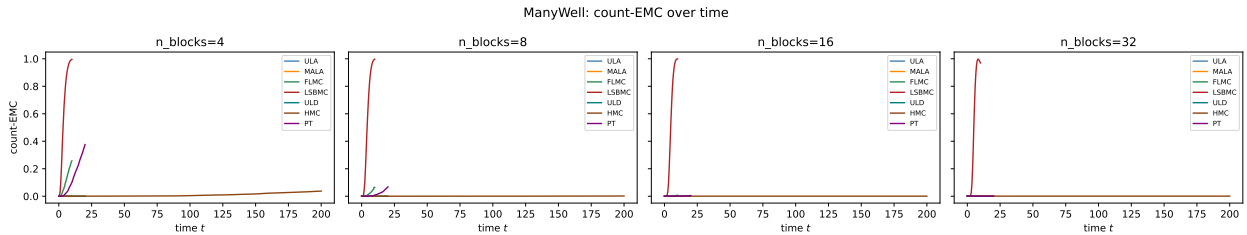


Figure 2: ManyWell count-EMC over sampling time (mean over seeds).

(LSB-MC) versus 1.41 (PT). PT starts respectably (count-EMC 0.38 at $d=8$) but decays toward zero, since getting all blocks right simultaneously is exponentially unlikely without directed jumps.

Discussion. Because the bank acts per block and brackets the separation, every barrier is crossed productively and LSB-MC pays essentially no penalty for added dimension, while undirected exploration decays as the modes multiply.

9 Experiment 3: a Bayesian posterior with a symmetry (label switching)

Setup. Inferring three exchangeable cluster means yields a posterior with six identical, label-permuted modes (§5); a correct sampler must weight all six equally.

Results. Table 5 reports the full metric set; Figure 3 tracks mode coverage over sampling time. Every baseline—including PT and HMC—stays in its initial mode (coverage 0.17, 1 of 6, EMC 0.011), whereas LSB-MC populates all six uniformly (coverage 1.00 = 6/6, EMC 0.998) and is closest in earth-mover distance ($W_2 = 7.82$ versus ≈ 14.63 for the trapped baselines).

Discussion. On a genuine posterior, exploring the symmetry is not optional: reporting one mode discards five-sixths of the answer. LSB-MC recovers the full posterior from a bank that needed no tuning beyond writing down the symmetry. As Remark 1 and §3 predict, discretization matters: an over-aggressive jump schedule strands particles between modes, and the configuration used here (finer Δt , more steps, moderate λ) clears that transit mass, returning the posterior-energy distribution to the reference’s low-energy band.

Table 3: ManyWell count-EMC across dimensions (mean $\pm$ SE).

Method	8	16	32	64
ULA	0.002 $\pm$ 0.000	0.002 $\pm$ 0.000	0.003 $\pm$ 0.000	0.004
MALA	0.001 $\pm$ 0.000	0.002 $\pm$ 0.000	0.003 $\pm$ 0.000	0.004
FLMC	0.258 $\pm$ 0.005	0.068 $\pm$ 0.002	0.013 $\pm$ 0.000	0.005 $\pm$ 0.000
LSBMC	0.995 $\pm$ 0.000	0.995 $\pm$ 0.001	0.995 $\pm$ 0.000	0.961 $\pm$ 0.001
ULD	0.001 $\pm$ 0.000	0.002 $\pm$ 0.000	0.003 $\pm$ 0.000	0.004
HMC	0.040 $\pm$ 0.003	0.005 $\pm$ 0.000	0.003 $\pm$ 0.000	0.004 $\pm$ 0.000
PT	0.376 $\pm$ 0.008	0.070 $\pm$ 0.005	0.010 $\pm$ 0.000	0.004 $\pm$ 0.000

Table 4: ManyWell results at $d = 64$ (mean $\pm$ SE over 5 seeds).

Method	Sliced W_2	MMD	count-EMC	count-KL	block-KL	time(s)	grads
ULA	2.083 $\pm$ 0.005	0.576 $\pm$ 0.001	0.004	5.515	3.000 $\pm$ 0.008	0.285	2000
MALA	2.130 $\pm$ 0.004	0.603 $\pm$ 0.001	0.004	5.515	3.937 $\pm$ 0.027	0.981 $\pm$ 0.000	4000
FLMC	1.062 $\pm$ 0.007	0.282 $\pm$ 0.001	0.005 $\pm$ 0.000	5.358 $\pm$ 0.006	0.335 $\pm$ 0.001	0.504	2000
LSBMC	0.178 $\pm$ 0.002	0.033 $\pm$ 0.000	0.961 $\pm$ 0.001	0.040 $\pm$ 0.001	0.001 $\pm$ 0.000	0.782	2000
ULD	2.140 $\pm$ 0.005	0.607 $\pm$ 0.001	0.004	5.515	4.654 $\pm$ 0.082	0.288	2001
HMC	1.545 $\pm$ 0.007	0.412 $\pm$ 0.001	0.004 $\pm$ 0.000	5.515 $\pm$ 0.000	0.856 $\pm$ 0.002	2.032	22000
PT	1.407 $\pm$ 0.014	0.375 $\pm$ 0.001	0.004 $\pm$ 0.000	5.464 $\pm$ 0.011	0.661 $\pm$ 0.005	1.619 $\pm$ 0.000	24000

10 Compute-fair comparison

A faster-mixing sampler is only interesting if the mixing is not bought with hidden compute. Table 6 reports, at the largest setting of each study, wall-clock time, gradient evaluations, and Lévy-score potential evaluations, with HMC charged $L+1$ gradients per proposal, PT charged T kernels per step, and LSB-MC charged its quadrature potential evaluations. Per-setting time and gradient counts also appear in the result tables of §7–§9.

Reading. LSB-MC’s cost is dominated by the quadrature and is problem-dependent. On ManyWell the correction is a 1-D grid lookup, so LSB-MC is the *cheapest* method (≈ 0.8 s at $d=64$ versus 1.6 s for PT) while being the most accurate—it wins on both axes. On MoG the correction integrates over a large nearest-neighbour bank, which is genuinely expensive at $K=80$ (≈ 184.5 s and ≈ 7.7 M potential evaluations, versus 2.6 s for PT): there the quality win carries a real per-step cost. The Bayesian model’s 513 s reflects the deliberately fine step size, not an intrinsic floor. The advantage is therefore not an artefact of extra compute, but on dense banks the quadrature is the price of admission.

11 Discussion

The three experiments vary, respectively, the mode count, the dimension, and the *origin* of the modes (a real symmetry). Across all three the ordering is stable: local methods stay put; PT helps but degrades as the problem grows; LSB-MC tracks the reference. The mechanism is the one made precise in §3: undirected crossings cost $\exp(2\Delta/\sigma^2)$, which compounds with mode count and dimension, whereas LSB-MC replaces them with directed jumps whose bias is cancelled exactly (Proposition 1), decoupling mixing from barrier height.

Bayesian GMM: mode coverage over time

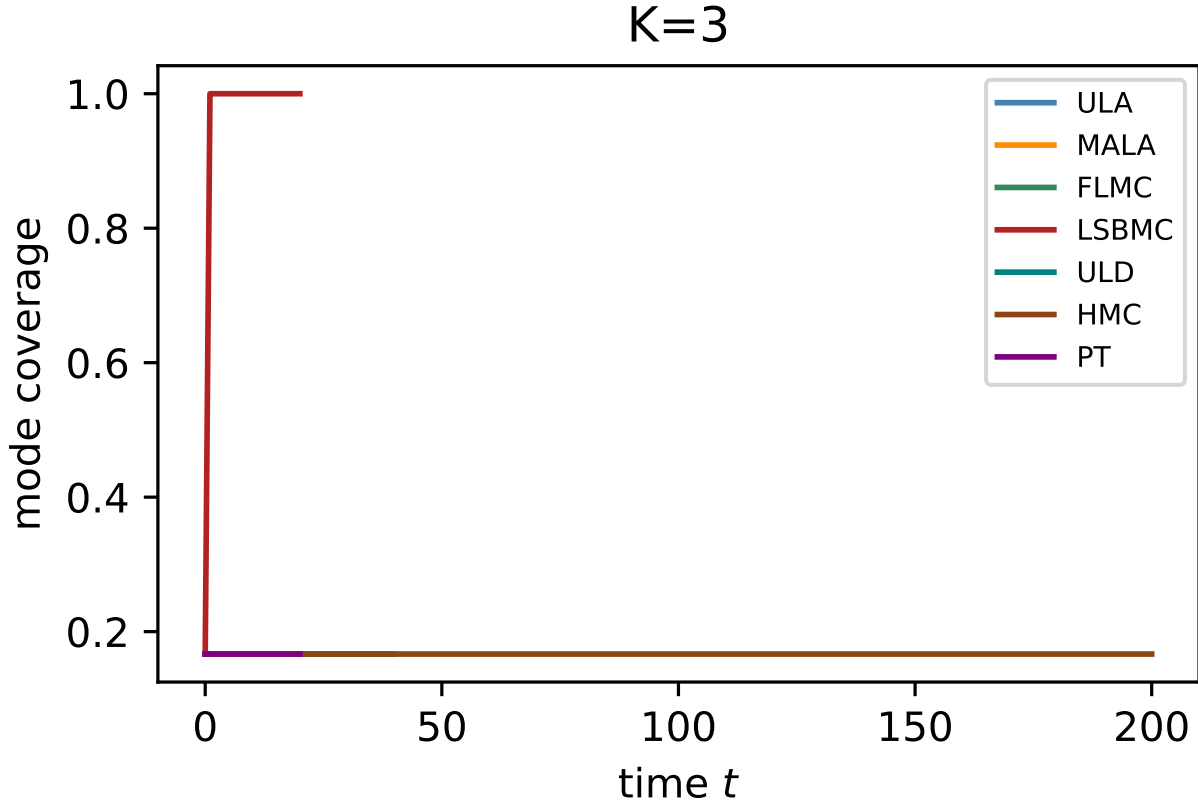


Figure 3: Bayesian GMM mode coverage over sampling time (mean over seeds).

12 Limitations

- **The bank needs structure.** ν is informed by the target geometry; this suits structured problems but is not black-box, whereas PT/HMC need no such input. Constructing good banks when the geometry is unknown is open.
- **No universal dominance.** Where a baseline is competitive the tables show it (e.g. FLMC on low- d ManyWell).
- **Dense banks cost compute** (§10).
- **Continuous-time theory, discrete implementation.** Proposition 1 is exact for $\mathcal{A}$; the Euler/quadrature scheme leaves an $O(\Delta t)$ bias and can strand particles if mis-tuned.
- **High- d W_2 is a proxy** (sliced- W_2).

Table 5: Bayesian GMM label-switching posterior ($K=3$, $\text{dim}=6$, 6 modes; mean $\pm$ SE over 5 seeds).

Method	cov.	#modes	EMC	KL	W_2	MMD	time(s)	grads
ULA	0.167	1	0.011	4.545	14.581 $\pm$ 0.066	0.654 $\pm$ 0.004	2.184	2000
MALA	0.167	1	0.011	4.545	14.627 $\pm$ 0.070	0.655 $\pm$ 0.004	5.615	4000
FLMC	0.167	1	0.011	4.545	14.610 $\pm$ 0.070	0.652 $\pm$ 0.004	1.900	2000
LSBMC	1.000	6	0.998 $\pm$ 0.000	0.002 $\pm$ 0.000	7.824 $\pm$ 0.684	0.040 $\pm$ 0.005	513.159	5000
ULD	0.167	1	0.011	4.545	14.630 $\pm$ 0.069	0.655 $\pm$ 0.004	1.850	2001
HMC	0.167	1	0.011	4.545	14.625 $\pm$ 0.069	0.655 $\pm$ 0.004	29.687	32000
PT	0.167	1	0.011	4.545	14.625 $\pm$ 0.067	0.655 $\pm$ 0.004	39.285	32000

13 Conclusion

Multimodal sampling is hard because modes are walled off and undirected steps pay an exponential price to cross. LSB-MC instead jumps along target-informed directions and corrects the induced bias exactly, so that π remains the invariant law (Proposition 1). Empirically, under strong baselines, interpretable metrics, two scaling axes, and a genuine Bayesian posterior, LSB-MC reaches and correctly weights essentially the whole distribution where the baselines stall—at a compute cost that is honestly reported and, on the high-dimensional benchmark, lower than the baselines’. The one requirement is a structure-informed jump bank, which is exactly what the structured, hardest-to-mix problems provide.

14 Recommendations for the main paper

Main paper. Replace the single-target showcase with two pieces of evidence that referees ask for: (i) a *scaling* result and (ii) a *realistic* benchmark.

- Promote the **MoG mode-count scaling** (Table 1) and the **ManyWell dimension-scaling** (Table 3) results to the main paper: they show how coverage / EMC degrade with K and d for the local baselines while LSB-MC stays close to the reference.
- Add the **Bayesian GMM label-switching** experiment (Table 5) as the headline “real model” result: the jump bank is derived from a genuine symmetry of the posterior rather than hand-tuned, which strengthens the methodological story.
- Include the strongest baselines (PT and HMC), not just ULA/MALA/FLMC. The honest comparison against PT is what makes the multimodal claim credible.
- Report the **compute-fairness** table (Table 6; wall-clock and gradient/potential evaluations) so the speed-up is not attributed to extra compute.

Appendix. Per-dimension and per- K tables, acceptance/swap diagnostics, the exact configurations, and the reference-construction details for the Bayesian posterior.

What the old section should become. The original double-well / single-MoG / single-ManyWell demonstrations are good *illustrations* and should move to the appendix or a “setup” figure; the quantitative claims should rest on the scaling studies and the Bayesian benchmark.

Table 6: Compute at the largest setting of each study (mean over 5 seeds): wall-clock time, gradient evaluations, and Lévy-score potential evaluations (millions). HMC is charged $L+1$ gradients per proposal, PT T kernels per step, LSB-MC its quadrature.

Method	time (s)	grad evals	pot. evals (M)
<i>MoG</i> , $K = 80$			
ULA	0.27	1500	0.000
MALA	0.86	3000	0.003
FLMC	0.43	1500	0.000
LSBMC	184.51	1500	7.680
ULD	0.26	1501	0.000
HMC	2.00	16500	0.003
PT	2.64	24000	0.026
<i>ManyWell</i> , $d = 64$			
ULA	0.28	2000	0.000
MALA	0.98	4000	0.004
FLMC	0.50	2000	0.000
LSBMC	0.78	2000	0.002
ULD	0.29	2001	0.000
HMC	2.03	22000	0.004
PT	1.62	24000	0.026
<i>Bayesian GMM</i> , $K = 3$			
ULA	2.18	2000	0.000
MALA	5.62	4000	0.004
FLMC	1.90	2000	0.000
LSBMC	513.16	5000	1.200
ULD	1.85	2001	0.000
HMC	29.69	32000	0.004
PT	39.28	32000	0.035

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A Configurations and hyperparameters

manywell scaling

- scaling: `n_blocks` $\in [4, 8, 16, 32]$
- methods: ULA, MALA, FLMC, LSBMC, ULD, HMC, PT
- particles 2048, steps 2000, dt 0.005, seeds [0, 1, 2, 3, 4]
- method configs: FLMC: {'alpha': 1.5, 'dt': 0.005}; HMC: {'dt': 0.05, 'n_leapfrog': 10}; LSBMC: {'dt': 0.005}; MALA: {'dt': 0.005}; PT: {'beta_min': 0.05, 'dt': 0.005, 'n_temps': 6, 'swap_interval': 5}; ULA: {'dt': 0.005}; ULD: {'dt': 0.01, 'friction': 2.0}
- target: {'jump_r_max': 3.8, 'jump_r_min': 3.2, 'lambda_block': 0.1}
- output: `results/iclr_sampling/20260628_032655_manywell_scaling`

mog scaling

- scaling: `n_modes` $\in [10, 20, 40, 80]$
- methods: ULA, MALA, FLMC, LSBMC, ULD, HMC, PT
- particles 1500, steps 1500, dt 0.01, seeds [0, 1, 2, 3, 4]
- method configs: FLMC: {'alpha': 1.5, 'dt': 0.01}; HMC: {'dt': 0.1, 'n_leapfrog': 10}; LSBMC: {'dt': 0.01, 'tame_cap': 10.0}; MALA: {'dt': 0.01}; PT: {'beta_min': 0.02, 'dt': 0.01, 'n_temps': 8, 'swap_interval': 5}; ULA: {'dt': 0.01}; ULD: {'dt': 0.02, 'friction': 2.0}
- target: {'center_range': 40.0, 'center_seed': 0, 'comp_sigma': 1.0, 'jump_chunk': 128, 'k_neighbors': 8, 'lam_total': 1.0, 'n_theta': 8}
- output: `results/iclr_sampling/20260628_032158_mog_scaling`

bayes gmm label switching

- scaling: `K` $\in [3]$
- methods: ULA, MALA, FLMC, LSBMC, ULD, HMC, PT
- particles 2000, steps 2000, dt 0.01, seeds [0, 1, 2, 3, 4]

Table 7: MoG results at $K = 10$ modes (mean $\pm$ SE over 5 seeds).

Method	W_2	MMD	cov.	EMC	KL	time(s)	grads
ULA	33.893 $\pm$ 0.634	0.623 $\pm$ 0.002	0.100	0.014 $\pm$ 0.000	4.286	0.367	1500
MALA	33.915 $\pm$ 0.652	0.623 $\pm$ 0.002	0.100	0.014 $\pm$ 0.000	4.286	0.882	3000
FLMC	33.525 $\pm$ 0.635	0.608 $\pm$ 0.003	0.200	0.021 $\pm$ 0.000	3.875 $\pm$ 0.002	0.435 $\pm$ 0.000	1500
LSBMC	17.526 $\pm$ 1.618	0.107 $\pm$ 0.006	1.000	0.924 $\pm$ 0.004	0.079 $\pm$ 0.004	4.401	1500
ULD	33.904 $\pm$ 0.630	0.623 $\pm$ 0.003	0.100	0.014 $\pm$ 0.000	4.286	0.276	1501
HMC	33.936 $\pm$ 0.611	0.623 $\pm$ 0.002	0.120 $\pm$ 0.020	0.014 $\pm$ 0.000	4.264 $\pm$ 0.022	2.125	16500
PT	33.591 $\pm$ 0.635	0.605 $\pm$ 0.004	0.200	0.021 $\pm$ 0.000	3.841 $\pm$ 0.004	1.156	24000

Table 8: MoG results at $K = 20$ modes (mean $\pm$ SE over 5 seeds).

Method	W_2	MMD	cov.	EMC	KL	time(s)	grads
ULA	36.033 $\pm$ 0.651	0.629 $\pm$ 0.004	0.090 $\pm$ 0.010	0.020 $\pm$ 0.000	3.933 $\pm$ 0.009	0.274	1500
MALA	36.069 $\pm$ 0.640	0.629 $\pm$ 0.004	0.090 $\pm$ 0.019	0.020 $\pm$ 0.000	3.937 $\pm$ 0.013	0.892	3000
FLMC	32.964 $\pm$ 0.733	0.549 $\pm$ 0.005	0.250	0.047 $\pm$ 0.000	3.058 $\pm$ 0.002	0.438	1500
LSBMC	11.008 $\pm$ 0.739	0.065 $\pm$ 0.005	1.000	0.937 $\pm$ 0.004	0.065 $\pm$ 0.004	15.150	1500
ULD	36.052 $\pm$ 0.641	0.629 $\pm$ 0.004	0.050	0.019	3.965	0.286	1501
HMC	35.754 $\pm$ 0.706	0.619 $\pm$ 0.004	0.210 $\pm$ 0.010	0.029 $\pm$ 0.000	3.544 $\pm$ 0.013	2.169	16500
PT	33.823 $\pm$ 0.747	0.564 $\pm$ 0.005	0.250	0.046 $\pm$ 0.000	3.078 $\pm$ 0.004	1.187	24000

- method configs: FLMC: {'alpha': 1.5, 'dt': 0.01}; HMC: {'dt': 0.05, 'n_leapfrog': 15}; LSBMC: {'dt': 0.004, 'metric_every': 250, 'n_steps': 5000}; MALA: {'dt': 0.01}; PT: {'beta_min': 0.02, 'dt': 0.01, 'n_temps': 8, 'swap_interval': 5}; ULA: {'dt': 0.01}; ULD: {'dt': 0.02, 'friction': 2.0}
- target: {'data_seed': 0, 'jump_chunk': 32, 'lam_total': 0.4, 'n_data': 300, 'n_theta': 8, 'p': 2, 'sigma_y': 1.0, 'tau': 10.0, 'true_radius': 6.0}
- output: results/iclr_sampling/20260628_031115_bayes_gmm_label_switching

B Additional per-setting tables

C Acceptance / swap diagnostics

Table 9: MoG results at $K = 80$ modes (mean $\pm$ SE over 5 seeds).

Method	W_2	MMD	cov.	EMC	KL	time(s)	grads
ULA	38.844 $\pm$ 0.411	0.675 $\pm$ 0.004	0.065 $\pm$ 0.002	0.065 $\pm$ 0.000	2.732 $\pm$ 0.002	0.270	1500
MALA	38.970 $\pm$ 0.501	0.676 $\pm$ 0.003	0.068 $\pm$ 0.003	0.065 $\pm$ 0.000	2.732 $\pm$ 0.003	0.862 $\pm$ 0.000	3000
FLMC	35.315 $\pm$ 0.512	0.594 $\pm$ 0.004	0.200 $\pm$ 0.007	0.094 $\pm$ 0.001	2.363 $\pm$ 0.005	0.434	1500
LSBMC	19.954 $\pm$ 0.663	0.180 $\pm$ 0.008	1.000	0.829 $\pm$ 0.005	0.188 $\pm$ 0.006	184.514	1500
ULD	39.243 $\pm$ 0.382	0.680 $\pm$ 0.003	0.062	0.063 $\pm$ 0.000	2.760 $\pm$ 0.002	0.264	1501
HMC	32.769 $\pm$ 0.364	0.547 $\pm$ 0.003	0.143 $\pm$ 0.003	0.094 $\pm$ 0.000	2.368 $\pm$ 0.003	1.997 $\pm$ 0.000	16500
PT	35.668 $\pm$ 0.573	0.604 $\pm$ 0.004	0.145 $\pm$ 0.005	0.087 $\pm$ 0.000	2.445 $\pm$ 0.005	2.638	24000

Table 10: ManyWell results at $d = 8$ (mean $\pm$ SE over 5 seeds).

Method	Sliced W_2	MMD	count-EMC	count-KL	block-KL	time(s)	grads
ULA	2.078 $\pm$ 0.009	0.627 $\pm$ 0.002	0.002 $\pm$ 0.000	6.441 $\pm$ 0.066	3.428 $\pm$ 0.030	0.405 $\pm$ 0.000	2000
MALA	2.086 $\pm$ 0.006	0.638 $\pm$ 0.002	0.001 $\pm$ 0.000	6.529 $\pm$ 0.023	3.920 $\pm$ 0.030	0.980 $\pm$ 0.000	4000
FLMC	1.057 $\pm$ 0.015	0.308 $\pm$ 0.002	0.258 $\pm$ 0.005	1.354 $\pm$ 0.018	0.339 $\pm$ 0.002	0.506	2000
LSBMC	0.202 $\pm$ 0.005	0.035 $\pm$ 0.001	0.995 $\pm$ 0.000	0.005 $\pm$ 0.000	0.001 $\pm$ 0.000	0.776	2000
ULD	2.097 $\pm$ 0.004	0.640 $\pm$ 0.002	0.001 $\pm$ 0.000	6.574 $\pm$ 0.001	4.437 $\pm$ 0.080	0.282	2001
HMC	1.525 $\pm$ 0.011	0.447 $\pm$ 0.002	0.040 $\pm$ 0.003	3.236 $\pm$ 0.061	0.834 $\pm$ 0.005	2.039	22000
PT	0.847 $\pm$ 0.017	0.260 $\pm$ 0.002	0.376 $\pm$ 0.008	0.978 $\pm$ 0.021	0.212 $\pm$ 0.003	1.313	24000

Table 11: ManyWell results at $d = 16$ (mean $\pm$ SE over 5 seeds).

Method	Sliced W_2	MMD	count-EMC	count-KL	block-KL	time(s)	grads
ULA	2.080 $\pm$ 0.010	0.603 $\pm$ 0.001	0.002 $\pm$ 0.000	6.247 $\pm$ 0.000	3.356 $\pm$ 0.019	0.283	2000
MALA	2.107 $\pm$ 0.006	0.617 $\pm$ 0.001	0.002 $\pm$ 0.000	6.248 $\pm$ 0.000	3.947 $\pm$ 0.039	1.002	4000
FLMC	1.060 $\pm$ 0.008	0.291 $\pm$ 0.002	0.068 $\pm$ 0.002	2.689 $\pm$ 0.034	0.339 $\pm$ 0.002	0.513	2000
LSBMC	0.178 $\pm$ 0.005	0.033 $\pm$ 0.001	0.995 $\pm$ 0.001	0.005 $\pm$ 0.001	0.001 $\pm$ 0.000	0.785	2000
ULD	2.118 $\pm$ 0.009	0.620 $\pm$ 0.001	0.002 $\pm$ 0.000	6.248 $\pm$ 0.000	4.608 $\pm$ 0.154	0.285	2001
HMC	1.525 $\pm$ 0.009	0.422 $\pm$ 0.002	0.005 $\pm$ 0.000	5.341 $\pm$ 0.064	0.835 $\pm$ 0.006	2.055	22000
PT	1.026 $\pm$ 0.025	0.292 $\pm$ 0.002	0.070 $\pm$ 0.005	2.665 $\pm$ 0.065	0.335 $\pm$ 0.002	1.285	24000

Table 12: ManyWell results at $d = 32$ (mean $\pm$ SE over 5 seeds).

Method	Sliced W_2	MMD	count-EMC	count-KL	block-KL	time(s)	grads
ULA	2.107 $\pm$ 0.006	0.590 $\pm$ 0.001	0.003 $\pm$ 0.000	5.871 $\pm$ 0.000	3.219 $\pm$ 0.007	0.280	2000
MALA	2.138 $\pm$ 0.006	0.608 $\pm$ 0.001	0.003 $\pm$ 0.000	5.871 $\pm$ 0.000	3.898 $\pm$ 0.047	0.975	4000
FLMC	1.056 $\pm$ 0.013	0.287 $\pm$ 0.001	0.013 $\pm$ 0.000	4.362 $\pm$ 0.026	0.336 $\pm$ 0.001	0.500	2000
LSBMC	0.174 $\pm$ 0.005	0.032 $\pm$ 0.000	0.995 $\pm$ 0.000	0.005 $\pm$ 0.000	0.000 $\pm$ 0.000	0.778	2000
ULD	2.149 $\pm$ 0.007	0.612 $\pm$ 0.001	0.003 $\pm$ 0.000	5.871 $\pm$ 0.000	4.568 $\pm$ 0.017	0.285	2001
HMC	1.550 $\pm$ 0.008	0.414 $\pm$ 0.001	0.003 $\pm$ 0.000	5.823 $\pm$ 0.002	0.838 $\pm$ 0.003	2.022	22000
PT	1.209 $\pm$ 0.005	0.327 $\pm$ 0.000	0.010 $\pm$ 0.000	4.661 $\pm$ 0.047	0.468 $\pm$ 0.002	1.266	24000

Table 13: Metropolis–Hastings acceptance and PT swap-acceptance rates at the largest setting of each study (mean over 5 seeds); “–” = not applicable to the method.

Method	MH acc.	PT swap
<i>MoG</i> , $K = 80$		
ULA	–	–
MALA	1.00	–
FLMC	–	–
LSBMC	–	–
ULD	–	–
HMC	1.00	–
PT	1.00	0.85
<i>ManyWell</i> , $d = 64$		
ULA	–	–
MALA	0.93	–
FLMC	–	–
LSBMC	–	–
ULD	–	–
HMC	0.89	–
PT	0.98	0.10
<i>Bayesian GMM</i> , $K = 3$		
ULA	–	–
MALA	0.77	–
FLMC	–	–
LSBMC	–	–
ULD	–	–
HMC	0.79	–
PT	0.95	0.52